# QUALITY INFORMATION DOCUMENT

## For
## OMI_CLIMATE_OHC_BALTIC_area_averaged_anomalies

**Issue: 2.0**

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# RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

| Issue | Date | § | Description of Change | Author | Validated By |
|---|---|---|---|---|---|
| 1.0 | 2024/09/01 | All | Creation of the document | U. Raudsepp<br>I. Maljutenko | P. Lagemaa |
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| | | | | | |

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# 1) INTRODUCTION

## a) Product Name

OMI_CLIMATE_OHC_BALTIC_area_averaged_anomalies.

## b) Indicator Evaluation Method

Ocean heat content anomaly is defined here as the deviation from a reference period (1993-2014) and is closely proportional to the average temperature change from z1 = 0 m to z2 = 300 m depth:

$$OHC = \int_{z_1}^{z_2} \rho_0 c_p (T_{mth} - T_{clim}) dz$$

,

with a spatially varying density ρ0 and specific heat capacity cp.

Averaged time series for ocean heat content and their error bars are calculated for the Baltic Sea.

## c) Copernicus Marine products used as input for the computation

| Product name (and Type) | Documentation |
|---|---|
| BALTICSEA_MULTIYEAR_PHY_003_011 (model reanalysis) | QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-011.pdf<br>PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BAL-PUM-003-011_012.pdf |

# 2) SCIENTIFIC VALIDATION

## a) Baltic Sea Ocean Freshwater Content

The method for calculating the ocean heat content anomaly is based on the daily mean sea water potential temperature fields (Tp) derived from the Baltic Sea reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011. The total heat content is determined using the following formula:

$$HC = \rho * cp * ( Tp + 273.15).$$

Here, ρ and cp represent spatially varying sea water density and specific heat, respectively, which are computed based on potential temperature, salinity and pressure using the UNESCO 1983 polynomial developed by Fofonoff and Millard (1983). The vertical integral is computed using the static cell vertical thicknesses sourced from the reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011 dataset cmems_mod_bal_phy_my_static, spanning from the sea surface to the ocean bottom. Spatial averaging is performed over the Baltic Sea spatial domain, defined as the region between 13° - 31° E and 53° - 66° N. To obtain the OHC annual anomaly time series in (J/m2), the mean heat content over the reference period of 1993-2014 was subtracted from the annual mean total heat content.

We evaluate the uncertainty from the mean annual error of the potential temperature compared to the observations from the Baltic Sea (Giorgetti et al., 2020). The uncertainty (shade, Figure 1) corresponds to the RMSD of the annual mean heat content biases (± 35.3 MJ/m²) evaluated from the observed temperatures and corresponding model values.

Linear trend (W/m2) has been estimated from the annual anomalies with the uncertainty of 1.96-times standard error.

More details and full scientific evaluation can be found in the Copernicus Marine Service Ocean State Report (Raudsepp et al., 2022).

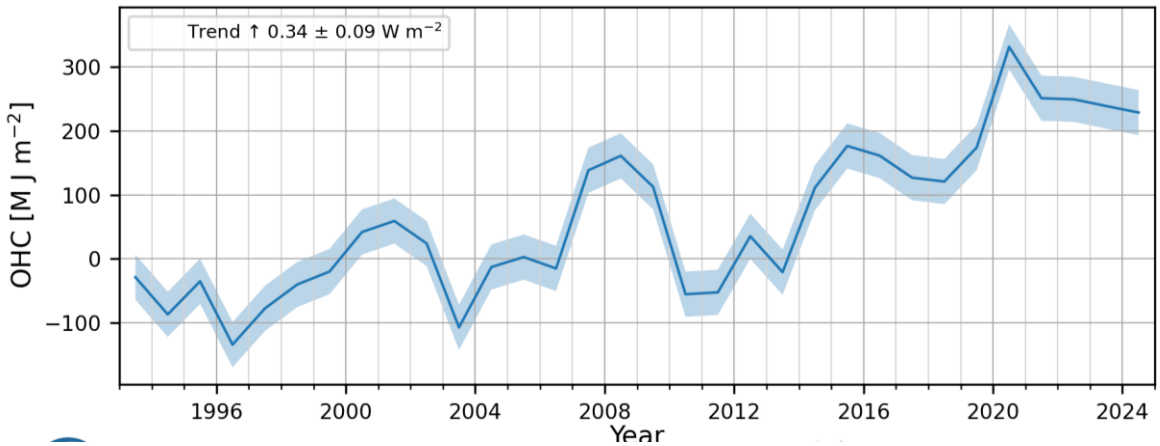


Figure 1. The time series of horizontally averaged ocean heat content anomaly integrated over 0-300 m depth, for the period of 1993-2024. The temperature from Copernicus Marine Service regional reanalysis product (BALTICSEA_MULTIYEAR_PHY_003_011) have been averaged over the Baltic Sea domain (13 °E - 31 °E; 53 °N - 66 °N; excluding the Skagerrak strait). The shaded area shows the estimated RMSD interval of annual heat content biases.

# 3) REFERENCES

Fofonoff, P. and Millard, R.C. Jr UNESCO 1983. Algorithms for computation of fundamental properties of seawater. UNESCO Tech. Pap. in Mar. Sci., No. 44, 53 pp., p.39. http://unesdoc.unesco.org/images/0005/000598/059832eb.pdf

Giorgetti, A., Lipizer, M., Molina Jack, M.E., Holdsworth, N., Jensen, H.M., Buga, L., Sarbu, G., Iona, A., Gatti, J., Larsen, M. and Fyrberg, L., 2020. Aggregated and Validated Datasets for the European Seas: The Contribution of EMODnet Chemistry. Frontiers in Marine Science, 7, p.583657.

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